TEXORA · LECTURE SHEET

Synthetic Fibres

The fibres polymerised from scratch — polyester, the nylons, acrylic, polypropylene and spandex: how each is made and spun, the three spinning methods side by side, and the property table that decides which fibre goes where.

TE-2108 · MAN-MADE TEXTILE FIBERS · TOPIC 3 · LEVEL 2 TERM 1

1. The three spinning methods

Every man-made fibre begins as a fluid — a melt or a solution — forced through a spinneret. What happens to the jet in the first metre decides the method's name:

- **Melt spinning** — the polymer is melted and extruded into cool air; the jet simply solidifies. Needs a polymer that melts without degrading: polyester, nylon, polypropylene. Cheapest and fastest (thousands of metres per minute), no solvent to recover.
- **Wet spinning** — the polymer solution (dope) is extruded into a liquid bath that extracts the solvent and precipitates/coagulates the polymer: viscose (regeneration bath), acrylic in aqueous baths, lyocell. Slow, but the only route when the polymer cannot melt.
- **Dry spinning** — the dope is extruded into hot air; the volatile solvent evaporates and is recovered: cellulose acetate in acetone, acrylic in DMF (some plants), spandex. Intermediate speed and cost.

In all three, the freshly solidified filament is then **drawn** — stretched typically 3–6 times its length between heated rollers or plates (for melt-spun yarns, often in a separate draw or draw-texturing step). Drawing aligns the chains along the axis, raises crystallinity, multiplies tenacity and cuts elongation to the designed value (TE-2107 Topic 1, §3).

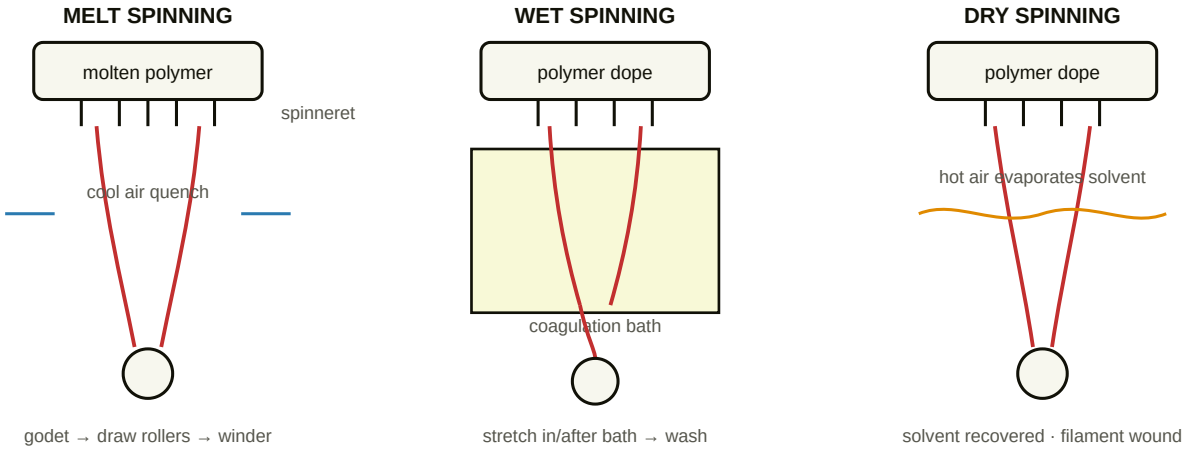


Figure 1 — The three spinning methods. Melt: the jet freezes in air. Wet: the jet precipitates in a bath. Dry: the solvent evaporates in hot air. Drawing follows in every case.

2. Polyester (PET)

Production outline: TPA (or DMT) + ethylene glycol → esterification/transesterification to BHET → melt polycondensation under vacuum at ≈ 280 °C → chips (or direct melt) → drying to very low moisture → **melt spinning** at 285–295 °C → quench air solidifies filaments → spin finish → winding as POY, then drawing/draw-texturing, or direct FDY; staple route cuts drawn tow.

Character: the balanced fibre — strong (≈ 4–6 g/den filament, higher for industrial yarns), resilient with outstanding crease recovery, dimensionally stable when heat-set, immune to moths and mildew, cheap on a vast scale. Weaknesses: regain 0.4 % (static, clammy in pure form for some apparel), disperse dyes only, melts and forms hard beads in flame. Roughly half of all world fibre production is polyester, much of it blended with cotton to marry polyester's strength and easy care to cotton's comfort.

3. Nylons

Production outline: nylon 6,6 — nylon salt solution concentrated, polymerised in autoclaves at ≈ 270–280 °C, extruded as ribbon/chips; nylon 6 — caprolactam ring-opening polymerisation at ≈ 250–270 °C, chips washed to remove residual monomer/oligomer. Both are **melt-spun** (≈ 260–295 °C), quenched, and drawn 3–5×.

Character: the toughness fibre — highest abrasion resistance of the common fibres, excellent elasticity and fatigue life, tenacity ≈ 4–7 g/den. Amide groups give 4 % regain and easy acid dyeing, but also acid sensitivity and the poorest UV resistance of the group. Uses follow the toughness: hosiery, carpets, tyre cord, ropes, parachutes, sportswear, lingerie.

4. Acrylic

Production outline: acrylonitrile copolymerised (≥ 85 % AN, with methyl acrylate and a dye-site comonomer — TE-2107 Topic 3) in aqueous or solvent systems → polymer dissolved to dope → **wet spinning** into aqueous coagulation baths, or **dry spinning** from DMF → washing, drawing in hot water/steam, drying (with controlled collapse and crimp for staple), crimping and cutting.

Character: the wool-substitute — bulky, warm, soft, with spun yarns that trap air like wool, at lower cost and with far better sunlight resistance and wash durability than wool; tenacity modest (≈ 2–3.5 g/den), regain ≈ 1.5–2 %, decomposes rather than melts. Knitwear, blankets, fleece, awnings; also the precursor fibre for carbon fibre.

5. Polypropylene and spandex

Polypropylene: isotactic PP polymerised with Ziegler–Natta catalysts → **melt-spun** at ≈ 230–280 °C, usually with pigment already in the melt (dope-dyed, since it cannot be dyed later). Lightest fibre (0.91 g/cm³ — it floats), zero regain, chemically inert, cheap; but melts at ≈ 165 °C, needs UV stabilisation, and holds oily stains. Carpets, upholstery, nonwovens, ropes, thermal base layers.

Spandex (elastane): segmented polyurethane block copolymer (TE-2107 Topic 1) → mostly **dry-spun** from DMF/DMAc. Only 2–5 % in a fabric transforms fit: 400–700 % extension with near-complete recovery. Chlorine and heat degrade it; it is always used covered or blended, never alone.

PROPERTY	POLYESTER	NYLON 6,6	ACRYLIC	POLYPROPYLENE
Density (g/cm³)	1.38	1.14	≈ 1.17	0.91
Tenacity (g/den)	4–6	4–7	2–3.5	4–6
Elongation at break (%)	15–30	20–40	20–35	20–40
Moisture regain (%)	0.4	4.0	1.5–2	≈ 0
Melting behaviour	Melts ≈ 258 °C	Melts ≈ 260 °C	Decomposes, no true melt	Melts ≈ 165 °C
Dyeing	Disperse only	Acid, disperse	Basic (cationic)	Dope-dyed only
Sunlight resistance	Good	Poorest of the group	Best of the group	Poor unless stabilised
Spinning method	Melt	Melt	Wet or dry	Melt

WORKED EXAMPLE — FILAMENT DIAMETER FROM DENIER

A polyester filament of 3.0 denier (PET density 1.38 g/cm³) — how thick is it?
Mass per centimetre = 3.0 g ÷ 900,000 cm = 3.33×10^{-6} g/cm.
Cross-sectional area = mass per cm ÷ density = $3.33 \times 10^{-6} \div 1.38 = 2.42 \times 10^{-6}$ cm².
Diameter = $2\sqrt{(A \div \pi)} = 2\sqrt{(2.42 \times 10^{-6} \div 3.1416)} \approx 1.75 \times 10^{-3}$ cm ≈ **17.5 µm**.
For comparison, a human hair is roughly 50–70 µm and a cotton fibre 12–20 µm wide — one reason fine-denier polyester can imitate natural-fibre hands, and why pushing below ≈ 1 dpf (about 10 µm in PET) earns the name microfibre.

6. Stress–strain behaviour — the family's fingerprints

Tenacity–elongation curves summarise each fibre's mechanical character: nylon stretches far and recovers (elastic region long), polyester is stiffer with a higher initial modulus, acrylic is weaker but bulky, and drawn industrial yarns trade elongation for tenacity. Reading these curves is the quickest way to predict sewing performance, fabric hand and end-use suitability (Figure 2 gives the shapes; values vary with draw ratio and test conditions).

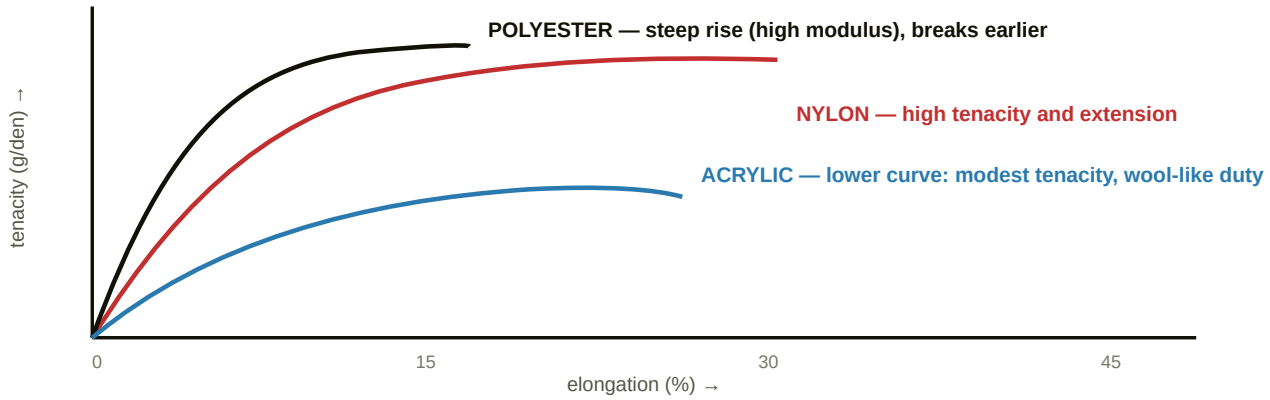


Figure 2 — Schematic tenacity–elongation curves (shapes representative, not measured data). Curve height = strength, length = extensibility, initial slope = modulus, area beneath = toughness (work of rupture).

Glossary

POY / FDY / DTY — partially oriented yarn; fully drawn yarn; draw-textured yarn. **Quench air** — the controlled cool-air stream that solidifies melt-spun filaments below the spinneret. **Draw ratio** — the ratio of output to input speed in drawing, equal to the length multiplication. **Dope-dyed (solution-dyed)** — pigmented in the melt/dope before extrusion. **Work of rupture** — the area under the stress–strain curve; the energy a fibre absorbs before breaking.

Key points

• Melt spinning serves polymers that melt safely (PET, nylon, PP); wet spinning serves those that cannot (viscose, much acrylic, lyocell); dry spinning serves volatile-solvent dopes (acetate, spandex, some acrylic).
• Drawing is universal: it creates orientation, crystallinity and tenacity in every family.
• Polyester = balance and scale; nylon = toughness and abrasion; acrylic = wool-like bulk with the best sunlight resistance; PP = lightest and cheapest but undyeable and low-melting; spandex = the stretch ingredient at 2–5 %.
• The property table's decisive columns in selection are regain, melting behaviour and dye class — they determine comfort, care and colouration before strength is even considered.

Practice

1. Match each fibre to its spinning method and justify in one clause: (a) cellulose acetate; (b) nylon 6; (c) acrylic (DMF plant); (d) polypropylene.
2. A 20 tex polyester yarn has tenacity 40 cN/tex. Calculate its breaking load in newtons.
3. Why must PET chips be dried immediately before melt spinning, while PP chips need no such care? Answer through the polymer backbones.
4. A buyer wants carpet yarn: high abrasion, solution-dyed, lowest cost per area covered. Choose between nylon and PP and defend the choice with two properties and one risk.
5. From Figure 2, which fibre has the highest work of rupture per unit mass, and which property of the curve are you using to judge?

Answer key

1. (a) Dry — acetate is spun from volatile acetone and cannot melt safely. (b) Melt — nylon 6 melts cleanly below its degradation range. (c) Dry (DMF evaporated in hot air) or wet in other plants — PAN decomposes before melting, so a solvent route is compulsory. (d) Melt — PP melts at ≈ 165 °C with a wide safe window.
2. Breaking load = 20 tex × 40 cN/tex = 800 cN = **8.0 N**.
3. PET's backbone contains ester links that hot water hydrolyses in the melt, cutting chains and IV (TE-2107 Topic 6). PP's backbone is pure C–C/C–H with no hydrolysable group and absorbs no water, so there is nothing for moisture to attack.
4. Polypropylene: lowest density (0.91) gives the most coverage per kilogram, and dope dyeing is cheap and completely fast; nylon would win on abrasion alone. Risk: PP's low melting point (≈ 165 °C) and poor resilience/UV unless stabilised — for a heavy-traffic contract carpet the nylon's abrasion and recovery may repay its higher price; state the traffic level that flips the decision.

5. Nylon. Work of rupture is the area under the tenacity–elongation curve, and nylon's curve encloses the largest area — high tenacity sustained over the longest extension.
